

Suricata and TCPDump

Comprehensive Documentation on Network Analysis and Threat Detection

Google Cybersecurity Professional Certificate

This documentation highlights the knowledge, practical experience, and technical skills acquired through the Google Cybersecurity Professional Certificate programme with a focus on **Suricata** and **TCPDump**. These tools are essential in modern cybersecurity operations for monitoring networks, analyzing packets, detecting suspicious activities, and supporting incident response processes.

Through hands-on practical exercises and real-world simulations, I developed a deeper understanding of how cybersecurity professionals use network analysis tools to investigate security events, monitor communications, and detect malicious activities within enterprise environments.

1. Introduction to Network Analysis

Network analysis is an important area in cybersecurity that involves monitoring, capturing, inspecting, and analyzing network traffic to identify suspicious or malicious activities. Organizations use network analysis tools to maintain visibility into communications across systems and to strengthen their overall security posture.

During the Google Cybersecurity programme, I learned how cybersecurity analysts investigate network traffic, inspect packets, identify anomalies, and use intrusion detection systems to monitor and protect networks.

2. Understanding Suricata

Suricata is an open-source intrusion detection and network monitoring system used for detecting threats, inspecting network traffic, and generating security alerts. The tool is widely used by cybersecurity analysts and Security Operations Centers (SOCs) to identify suspicious activities within networks.

Through the programme, I learned how Suricata performs deep packet inspection and how custom detection rules can be created to monitor specific network behaviors.

3. Practical Skills Learned with Suricata

The practical labs provided hands-on exposure to configuring and using Suricata for network monitoring and threat detection.

Skills developed included:

- Creating and modifying custom Suricata rules
- Analyzing packet capture (PCAP) files
- Detecting suspicious HTTP requests
- Monitoring TCP network communications
- Investigating generated alerts
- Reviewing Suricata log files
- Interpreting JSON-formatted event logs

The exercises improved my understanding of how intrusion detection systems identify and report suspicious activities in real-world environments.

4. Suricata Log Analysis

A major aspect of the course involved analyzing Suricata-generated logs. I learned how analysts use these logs to investigate incidents and understand network events.

The primary log files analyzed included:

- **fast.log** – Contains summarized alert information
- **eve.json** – Stores detailed JSON-formatted event data
- **suricata.log** – Contains engine and runtime information

Using Linux command-line tools, I practiced reviewing these logs to identify suspicious network activities and security alerts.

5. Understanding TCPDump

TCPDump is a command-line packet analyzer used to capture and inspect network traffic in real time. It is commonly used by cybersecurity analysts and network administrators for troubleshooting, monitoring communications, and investigating suspicious activities.

Through the Google Cybersecurity programme, I learned how TCPDump captures packets traversing a network interface and how the captured traffic can be analyzed for incident investigation and threat detection purposes.

6. Practical Skills Learned with TCPDump

Hands-on exercises using TCPDump improved my understanding of packet-level network analysis.

Skills practiced included:

- Capturing live network traffic
- Filtering traffic based on protocols and ports

- Inspecting TCP and HTTP communications
- Monitoring source and destination IP addresses
- Investigating suspicious traffic patterns
- Saving packet captures for later analysis

These exercises demonstrated how analysts use TCPDump to investigate network behavior and identify malicious activities.

7. Linux Command-Line Operations

The programme also strengthened my Linux command-line skills, which are essential for cybersecurity investigations and network analysis.

Commands practiced during the labs included:

```
# Run Suricata against a packet capture file
sudo suricata -r sample.pcap -S custom.rules -k none
```

```
# View Suricata logs
cat /var/log/suricata/fast.log
```

```
cat /var/log/suricata/eve.json
```

```
# Format JSON output
jq . /var/log/suricata/eve.json | less
```

```
# Capture packets with TCPDump
sudo tcpdump -i eth0
```

```
# Capture HTTP traffic
sudo tcpdump -i eth0 port 80
```

```
# Save captured traffic to a file
sudo tcpdump -w capture.pcap
```

These command-line exercises improved my ability to inspect network communications, analyze captured traffic, and investigate alerts generated from suspicious activities.

8. Suricata and TCPDump in Network Analysis

The course demonstrated how Suricata and TCPDump complement each other in cybersecurity operations.

TCPDump is primarily used for packet capturing and traffic inspection, while Suricata provides advanced intrusion detection and alert generation based on predefined rules and signatures.

Together, these tools support:

- Network traffic monitoring
- Threat detection and analysis
- Security event investigation
- Intrusion detection
- Incident response activities
- Packet-level forensic analysis

Understanding how both tools operate together strengthened my knowledge of real-world network security monitoring workflows.

9. Practical Learning Experience

The Google Cybersecurity Professional Certificate programme provided practical labs and simulations that enhanced my understanding of network security operations. The hands-on activities enabled me to work directly with cybersecurity tools used by professionals in the industry.

Through these exercises, I developed confidence in:

- Investigating suspicious network activity
- Reviewing intrusion detection alerts
- Analyzing network packets
- Understanding communication protocols
- Using Linux-based cybersecurity tools
- Performing security monitoring tasks

The practical experience significantly improved my technical knowledge and analytical thinking in cybersecurity operations.

10. Conclusion

The Suricata and TCPDump practical exercises within the Google Cybersecurity Professional Certificate programme provided valuable real-world cybersecurity experience. Through hands-on learning, I gained important technical skills in network traffic analysis, packet inspection, intrusion detection, log analysis, and security monitoring.

The programme strengthened my understanding of how cybersecurity professionals detect threats, investigate suspicious communications, and respond to security incidents using industry-standard tools.

Overall, the training prepared me with foundational and practical cybersecurity skills relevant to network analysis, security operations, and incident response environments.